

Part 13: Rotation

1. **What is rotation in 8086?** It shifts bits left or right, but unlike standard shifts, bits falling off one end are put back onto the other end or into the carry flag.
2. **How does ROL work? Rotate Left:** Shifts bits left; the MSB moves into the LSB position *and* the Carry Flag.
3. **How does ROR work? Rotate Right:** Shifts bits right; the LSB moves into the MSB position *and* the Carry Flag.
4. **How does RCL differ from ROL? Rotate Carry Left:** It treats the Carry Flag as part of the circle, rotating bits through the Carry Flag (9 bits for byte, 17 bits for word).
5. **How does RCR differ from ROR? Rotate Carry Right:** It rotates bits through the Carry Flag, acting as an extension of the register.
6. **How do carry flags affect rotation?** In `RCL / RCR`, the Carry Flag is part of the loop. In `ROL / ROR`, the Carry Flag just stores a copy of the bit that was rotated.
7. **How do you rotate multiple bits?** Move the count into `CL` (e.g., `MOV CL, 4`) and then use the instruction (e.g., `ROL AX, CL`).
8. **How do you use rotation for encryption?** Rotating bits rearranges them in a way that scrambles data, which can be part of a simple encryption algorithm.
9. **How do you rotate a byte stored in memory?** Use the `PTR` directive to specify size. Example: `ROL BYTE PTR [SI], 1`.
10. **How do you rotate 16-bit numbers?** Use a 16-bit register like `AX`. Example: `ROR AX, 1`.
11. **How is rotation different from shift?** Shift instructions throw away bits that fall off (or store them only in Carry), while rotation instructions recycle them back into the number.
12. **How do you rotate circularly?** `ROL` and `ROR` are circular rotations by definition—no bits are lost.
13. **How do you rotate signed numbers?** Rotation treats data as raw bits, so it ignores signs. You just rotate the bit pattern regardless of what it represents.
14. **How do rotations affect high and low bits?** In `ROL`, high bits wrap around to low positions. In `ROR`, low bits wrap around to high positions.
15. **How do you use rotation in masking operations?** You can rotate a bit of interest into the LSB or MSB position to easily test it or isolate it.